

☒ 40 MCQs with Answers & Explanations

☒ Easy (10 Questions)

Q1. What is pfSense primarily used for?

- A) Operating system development
- B) Network simulation
- C) Firewall and router
- D) Virtualization

Answer: C

Q2. Which protocol is typically used in VPN configuration in pfSense?

- A) HTTP

- B) FTP
- C) OpenVPN
- D) SMTP

Answer: C

Q3. ClearOS is based on which Linux distribution?

- A) Ubuntu
- B) Fedora
- C) CentOS
- D) Debian

Answer: C

Q4. What is the role of a **reverse proxy**?

- A) Routes internal traffic out
- B) Blocks internet access
- C) Handles client requests on behalf of backend servers
- D) Encrypts local databases

Answer: C

Q5. Which tool is best known for **web caching** and can act as a reverse proxy?

- A) iptables
- B) Squid

- C) MySQL
- D) Apache

Answer: B

Q6. What does UTM stand for?

- A) Unified Traffic Manager
- B) Uniform Threat Manager
- C) Unified Threat Management
- D) User Threat Monitor

Answer: C

Q7. Which Nginx feature enables it to serve multiple backend servers?

- A) Modules
- B) Upstreams
- C) Proxies
- D) Threads

Answer: B

Q8. Server load balancing is used to:

- A) Encrypt data
- B) Format hard disks
- C) Distribute client requests among servers

D) Update Linux kernels

Answer: C

Q9. Which of the following is **not** a typical UTM function?

A) IDS/IPS

B) Email filtering

C) Firewall

D) Text editor

Answer: D

Q10. In Nginx, what directive defines the server to forward requests to?

A) server_root

B) proxy_pass

C) backend_address

D) listen_proxy

Answer: B

☒ **Next:** I'll continue with the **20 medium** and **10 hard** level MCQs in the following message. Ready?

You said:

yes

ChatGPT said:

Great! Here's the continuation:

☒ **Linux Software Firewall, Reverse Proxy, UTM, Load Balancing – MCQs (continued)**

☒ **Medium (20 Questions)**

Q11. What feature in pfSense allows filtering based on **application signatures**?

- A) NAT
- B) VPN
- C) Snort IDS/IPS
- D) OpenDNS

Answer: C

☒ *Snort plugin in pfSense provides signature-based intrusion detection.*

Q12. Which of the following is a **true benefit of ClearOS**?

- A) Needs command-line configuration only
- B) No GUI
- C) Combines UTM features in a modular web interface
- D) Requires Windows OS

Answer: C

Q13. Which command-line tool does pfSense primarily use in the background for firewall rules?

- A) iptables
- B) nftables
- C) pf (Packet Filter)
- D) ufw

Answer: C

☒ *pfSense is built on FreeBSD, which uses PF instead of iptables.*

Q14. Nginx load balancing supports which of the following?

- A) CPU-based routing
- B) Geolocation blocking
- C) Least connections algorithm
- D) Email encryption

Answer: C

Q15. Squid reverse proxy is typically used to:

- A) Host virtual machines
- B) Cache and forward HTTP requests to backend servers
- C) Manage databases
- D) Serve as a DHCP server

Answer: B

Q16. In Nginx, to serve traffic to multiple servers in rotation, we use:

- A) FastCGI
- B) proxy_pass with loopback
- C) upstream block
- D) SSL termination

Answer: C

Q17. What is the primary difference between forward and reverse proxy?

- A) Forward proxies serve backend servers
- B) Reverse proxies serve clients on behalf of backend
- C) Reverse proxies use more encryption
- D) Forward proxies handle TCP only

Answer: B

Q18. What pfSense feature allows mapping external to internal ports?

- A) IPsec VPN
- B) NAT port forwarding
- C) Load balancing
- D) Traffic shaping

Answer: B

Q19. Which load balancing technique sends traffic to the server with the **fewest connections**?

- A) Round Robin
- B) Random
- C) Least Connections
- D) Hashing

Answer: C

Q20. Which plugin in pfSense helps **block ads and malicious domains**?

- A) Squid
- B) pfBlockerNG
- C) ClamAV
- D) Suricata

Answer: B

Q21. What is the function of cache_peer in Squid reverse proxy configuration?

- A) Encrypt the cache
- B) Define backend servers
- C) Schedule cron jobs
- D) Block HTTPS

Answer: B

Q22. Which UTM component scans email attachments for malware?

- A) IDS
- B) Anti-spam engine
- C) IPsec
- D) DHCP

Answer: B

Q23. Nginx can terminate SSL connections, meaning:

- A) It encrypts all backend connections
- B) It decrypts SSL and sends plain HTTP to backend
- C) It blocks non-SSL traffic
- D) It prevents VPN tunneling

Answer: B

Q24. pfSense traffic shaping allows:

- A) Disk formatting
- B) Prioritizing certain traffic like VoIP
- C) Blocking MAC addresses only
- D) NAT loopback

Answer: B

Q25. ClearOS web proxy is typically used to:

- A) Filter kernel modules
- B) Encrypt VoIP calls
- C) Control internet usage and cache HTTP
- D) Host cloud storage

Answer: C

Q26. Which of these is **not** part of a typical UTM device?

- A) Content filtering
- B) Web hosting service
- C) VPN gateway
- D) IDS/IPS

Answer: B

Q27. What does HAProxy specialize in?

- A) DNS filtering
- B) IP spoofing
- C) High-performance load balancing
- D) File encryption

Answer: C

Q28. A reverse proxy provides which **security benefit**?

- A) Hides backend server IPs from clients
- B) Encrypts local file systems
- C) Prevents disk failures
- D) Encrypts DNS requests

Answer: A

Q29. What is a potential **downside** of putting all services into one UTM device?

- A) Too many ports

- B) Lower CPU usage
- C) Single point of failure
- D) Requires cloud integration

Answer: C

Q30. Which pfSense feature detects packet anomalies and payload-based attacks?

- A) NAT
- B) Suricata or Snort IDS
- C) Port forwarding
- D) Gateway groups

Answer: B

☒ Hard (10 Questions)

Q31. In Nginx, what does the directive keepalive in upstream block control?

- A) DNS caching time
- B) Backend connection reuse
- C) Number of retries
- D) Client IP hashing

Answer: B

Q32. What kind of UTM-based attack detection relies on **heuristics and behavior patterns**?

- A) Signature-based IDS
- B) Anomaly-based IDS
- C) Port scanning
- D) Whitelisting

Answer: B

Q33. How does least_conn improve over round robin in load balancing?

- A) Prevents IP conflicts
- B) Ensures client IP persistence
- C) Avoids overloading busy servers
- D) Encrypts backend communication

Answer: C

Q34. What is a drawback of reverse proxy caching in dynamic content?

- A) It slows down static pages
- B) Cache can serve stale data if not configured properly
- C) Increases server CPU
- D) Requires kernel changes

Answer: B

Q35. Which component in UTM provides **gateway-level antivirus scanning**?

- A) Squid
- B) Suricata
- C) ClamAV
- D) Snort

Answer: C

Q36. Which pfSense feature balances **multiple WANs** for failover?

- A) Firewall alias
- B) Gateway groups
- C) IPsec tunnels
- D) NAT reflection

Answer: B

Q37. What does `originserver` in Squid's `cache_peer` directive mean?

- A) Peer acts as primary web server
- B) Enables authentication
- C) Caches forwarded requests
- D) Blocks POST requests

Answer: A

Q38. What is the biggest risk if a UTM device is **misconfigured**?

- A) Slow boot time
- B) OS crash
- C) Creates a single point of failure
- D) Hardware overclocking

Answer: C

Q39. Which load balancing method maps specific clients to backend servers using consistent values?

- A) Round robin
- B) Least connection
- C) IP hash
- D) Time slicing

Answer: C

Q40. How can Nginx be used in **active health checks** of backend servers?

- A) Using proxy_redirect
- B) Using http_stub_status_module
- C) Using health_check directive with third-party module
- D) Using listen default_server

Answer: C

